Simulated Land use and land cover change under scenarios of Ecological Infrastructure development in Switzerland between 2020-2060

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Abstract

This is the abstract.

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1. Introduction

The mountainous central European country of Switzerland, like the majority of countries globally, exhibits declining biodiversity due to multiple drivers such as urbanization, climatic change and environmentally deleterious agricultural practices. To address biodiversity loss, in 2012, the federal government of Switzerland ratified the Swiss Biodiversity Strategy (SBS)(**?**, foen)2012). One of the goals of the SBS is the development of Switzerland's Ecological Infrastructure (EI) as a "network of protected and connected areas that contribute to safeguarding the essential services of ecosystems for society and economy … and form the basis for a rich and resilient biodiversity" (**?**, foen)2023). Achieving this goal will require consideration of not just how Switzerland's EI should be planned under existing conditions but how it's development in space and time could proceed in concurrence with changes in the boundary conditions of the socio-ecological system within which it is situated.

In this regard, Mayer et al. (2023) defined 5 narrative scenarios to frame the development of EI in Switzerland between 2020 and 2060 alongside expected demographic, economic and climatic changes in Switzerland. The scenarios were

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produced through a participatory process with the aim of envisioning nature-positive outcomes using the IPBES Nature futures framework. As such the scenarios can characterized as both exploratory and normative, encapsulating the development of a "functioning" EI from different perspectives of societal-nature relationships (ibid).

However, whilst descriptive scenarios are a useful starting point for planners and decision-makers, their legitimacy can be increased by operationalising the narratives through computer-based simulation modelling so that their spatio-temporal impacts on critical aspects of the study system can be visualized and explored in detail (Verburg et al., 2004). Such operationalization of scenarios is especially desirable when their realization would likely result in trade-offs that must be negotiated by society. This is pertinent in the case of EI, which, under the definition employed by the SBS, aims to balance both ecosystem service ES) provision and biodiversity protection, which have often been acknowledged as discordant objectives.

In terms of operationalizing the specific scenarios of EI development from Mayer et al. (2023), the proximal linkages between EI as a 'network of areas' at the landscape level in Switzerland and it's intended outcomes for biodiversity and ES are difficult to model directly. Instead, it is logical to focus on a higher level of abstraction and model an intermediary, or distal, interaction between the physical expression of EI and these outcomes. A common component in the modelling of both measures of biodiversity (e.g species distribution) and ES provision (Newbold et al., 2015; Gomes et al., 2021) is land use and land cover (LULC). Furthermore LULC change (LULCC) trends are also a central theme in the SBS concept of EI as well as the scenario narrative's of Mayer et al. (2023). In particular, the role of protected areas and Other Effective Conservation Measures (OECMs) as components of the EI as well as other EI-relevant LULCC trends such as urban sprawl and the spatial distribution of agricultural production. On this basis, the goal of this research is to operationalize the scenario's of EI development produced for Switzerland by Mayer et al. (2023) by simulating their implications for trends in LULCC in Switzerland between 2020 and 2060 using a temporally dynamic Constrained Cellular Automata (CCA) model.

CCA models have enjoyed enduring popularity for the modelling of scenarios of LULCC for the last three decades. LULCC-CCAs treat landscapes as a pixelated grid of cells of individual LULC states with cells changing state over time typically based on two processes. Firstly, the spatial distribution of potential LULCC is determined through calculation of cellular probabilities of change, often referred to as transition potential (TP), to alternate LULC states. The method by which this is achieved, differs between popular LULCC-CAs although it is commonly based on the quantification of correlative relationships with explanatory variables such as biophysical, socio-economic and neighbourhood (focal LULC) predictors using a statistical model (García-Álvarez et al., 2022). Secondly, the amount of cellular LULC changes that occur in a given

simulation time step is prescribed by values of transition rates extrapolated from historic LULC data, with an allocation algorithm selecting the required number of cells to change based upon their transition probabilities .

Considering the two processes described above LULCC-CCAs offers several opportunities, or mechanisms, to operationalize aspects of scenarios. Firstly, the data supplied in the form of dynamic variables used to predict cellular TP values can be specific to the different scenario's being modelled. For example, in scenarios contrasting differential levels of climatic change, the data provided may come from exogenous climatic models of different Relative Concentration Pathways(Song et al., 2019). A second mechanism is to alter predicted cellular TP values according to proscribed criteria, this can be done directly: such as spatially designated land management policies, for example reducing the potential for transition to urban LULC states inside environmental protection areas(Gharbia et al., 2016; Price et al., 2017; Clerici et al., 2019; Daneshi et al., 2021; Peng et al., 2021; Frank et al., 2023). Or indirectly by altering the influence of variables within the model used to calculate TPs(Troupin and Carmel, 2016; Gago-Silva et al., 2017). Another means is to adjust parameters of the LULCC-CCA's allocation algorithm, that may for example increase the creation of new patches of LULC class cover relative to the expansion of existing areas (Prieto-Amparán et al., 2019). Finally, the influence of different scenarios can be incorporated through specification of the transitions rates (Shrestha et al., 2018; Cheng et al., 2020; Schirpke et al., 2020; Peng et al., 2021).

The influence of these mechanisms within a generalized schematic of an LULCC-CCA model are highlighted visually in Figure 1 below. Many LULCC-CCA scenario simulation studies utilize a combination of these mechanisms, however it is rare that they are all implemented in a detailed fashion.

To perform scenario-based simulation of future LULCC in Switzerland between 2020 and 2060 we utilized a constrained cellular automata (CCA) modelling approach combining R (4.2.2)(R core team, 2022) with the environment modelling software Dinamica EGO (7.2.0)(Soares-Filho et al., 2009).

2. Methods

2.1. Model overview

Figure 1 highlights the primary processes taking place within the LULCC-CCA model, the details of each these processes are summarized in the subsequent sections although for brevity some material has been presented as appendices.

2.2. Land use land cover transition identification

The process of identifying LULC transitions, i.e. pixel changes from an initial specific LULC class to a different final LULC class, was based upon a preceding

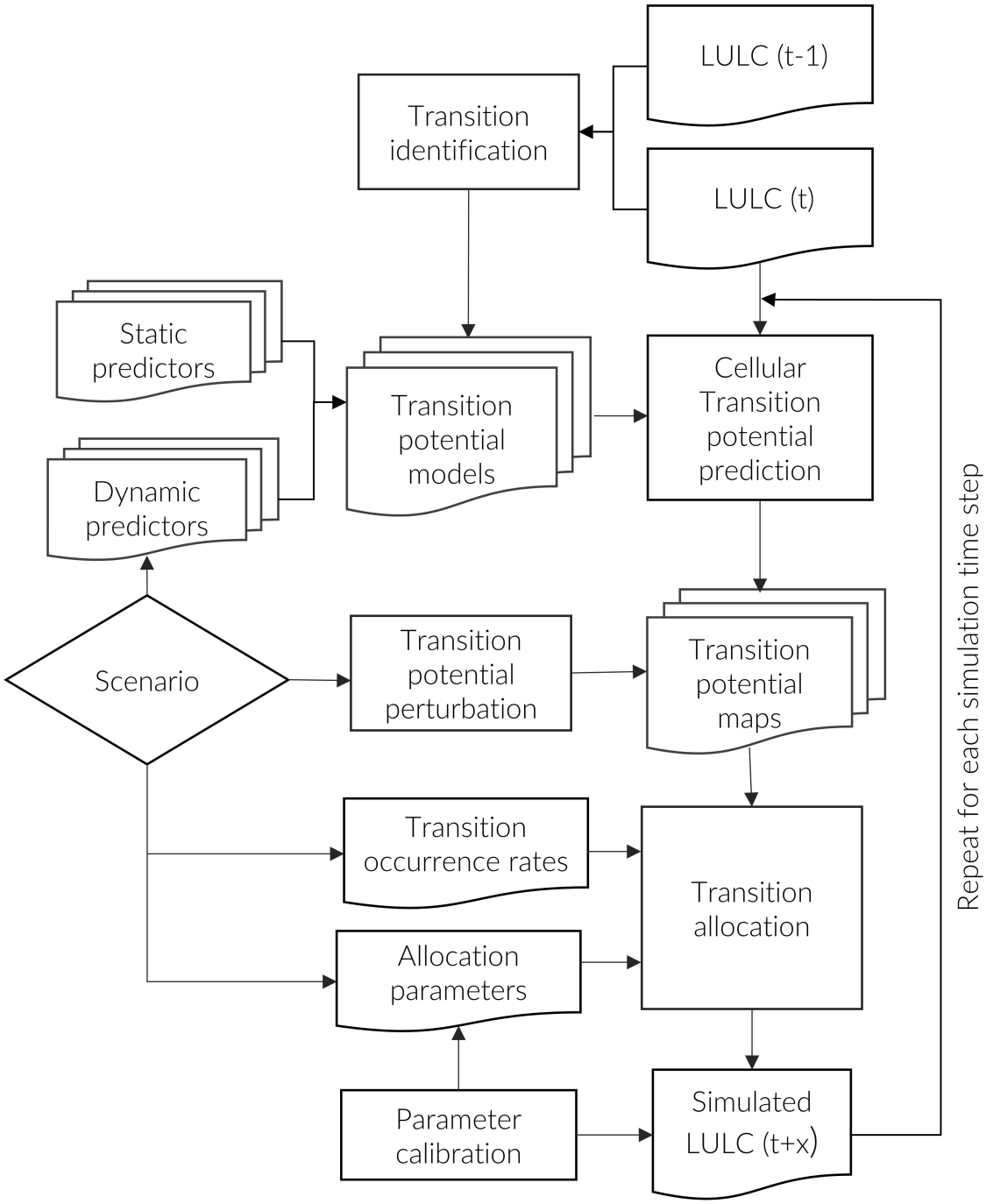

Figure 1: test

study by Black et al. (2023). We utilized classified LULC datasets for four time periods from the Swiss area statistics (SFSO, 2021): 1981-1985; 1992-1997; 2004-2009; 2013-2018 (Note that date range is the extent of the data-collection flying period, and in fact each product represents a single LULC map). Given that transitions are identified between two time points these four historic LULC maps meant that we had three LULC transition periods, which will henceforth be referred to using the latter year of the flying period as: 1985-1997; 1997-2009; and 2009-2018 respectively. Following Black et al. (2023) we aggregated the original 72 LULC classes in the data to 10 classes: Alpine pastures; Closed forest; Glacier; Grassland; Intensive agriculture; Open Forest; Shrubland; Permanent crops; Settlement/urban/amenities; Static (see Appendix A for a table of class aggregations)

2.3. Transition potential modelling

2.4. CA parameter calibration

2.5. Scenario operationalisation

2.5.1. Transition rates

Price et al. 2017 use a similar process as we plan to calculate the expected increases in the 'most important' categories and then calculate increates in the other categories from the remaining land.

Price et al. 2017 on calculating demand for urban area: "To calculate a total urban area demand per scenario, we used the modelled population level for 2060 in combination with the minimum and maximum of current per capita urban area demand data for the five cantons in our study area."

2.6. Validation

2.7. Simulation

3. Results

4. Discussion

5. Figures and tables

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